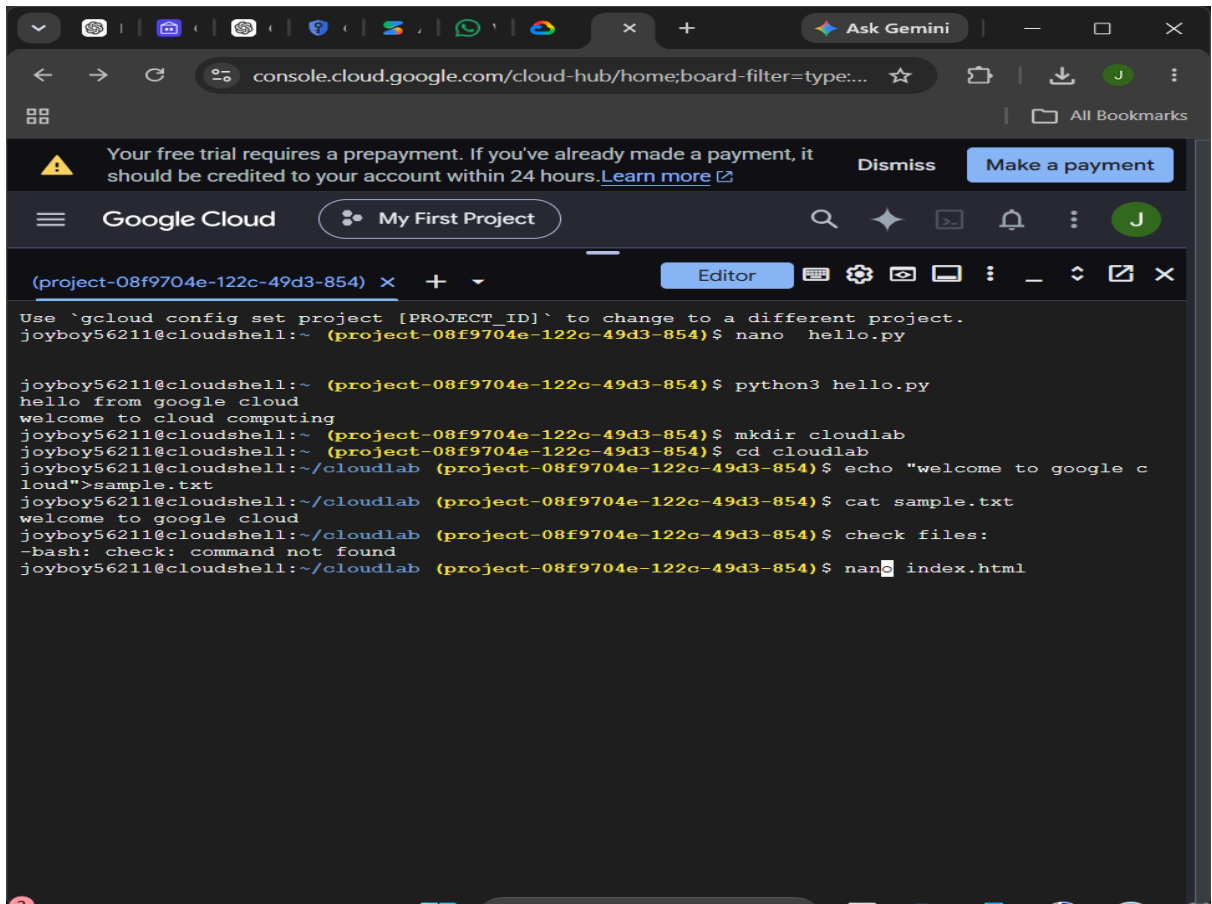


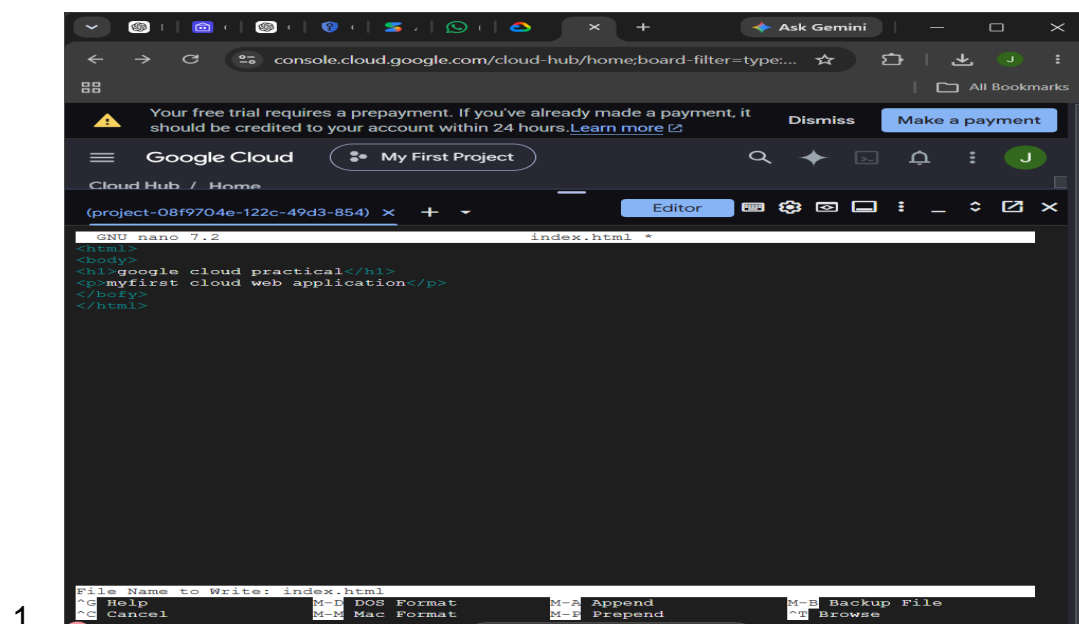
## 1. Execute a Python Program in Google and 2. Create and Manage Files in Cloud Storage of Cloud Shell



The screenshot shows the Google Cloud Shell interface. At the top, there's a navigation bar with the Google Cloud logo and 'My First Project'. Below it, a terminal window is open for project-08f9704e-122c-49d3-854. The terminal shows the following commands and output:

```
(project-08f9704e-122c-49d3-854) $ nano hello.py
joyboy56211@cloudshell:~ (project-08f9704e-122c-49d3-854) $ python3 hello.py
hello from google cloud
welcome to cloud computing
joyboy56211@cloudshell:~ (project-08f9704e-122c-49d3-854) $ mkdir cloudlab
joyboy56211@cloudshell:~ (project-08f9704e-122c-49d3-854) $ cd cloudlab
joyboy56211@cloudshell:~/cloudlab (project-08f9704e-122c-49d3-854) $ echo "welcome to google c
loud">sample.txt
joyboy56211@cloudshell:~/cloudlab (project-08f9704e-122c-49d3-854) $ cat sample.txt
welcome to google cloud
joyboy56211@cloudshell:~/cloudlab (project-08f9704e-122c-49d3-854) $ check files:
-bash: check: command not found
joyboy56211@cloudshell:~/cloudlab (project-08f9704e-122c-49d3-854) $ nano index.html
```

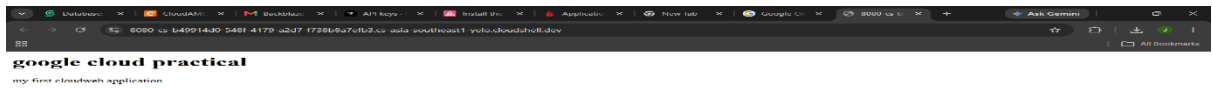
## 3. Run a Simple Web Server in Google Cloud Shell



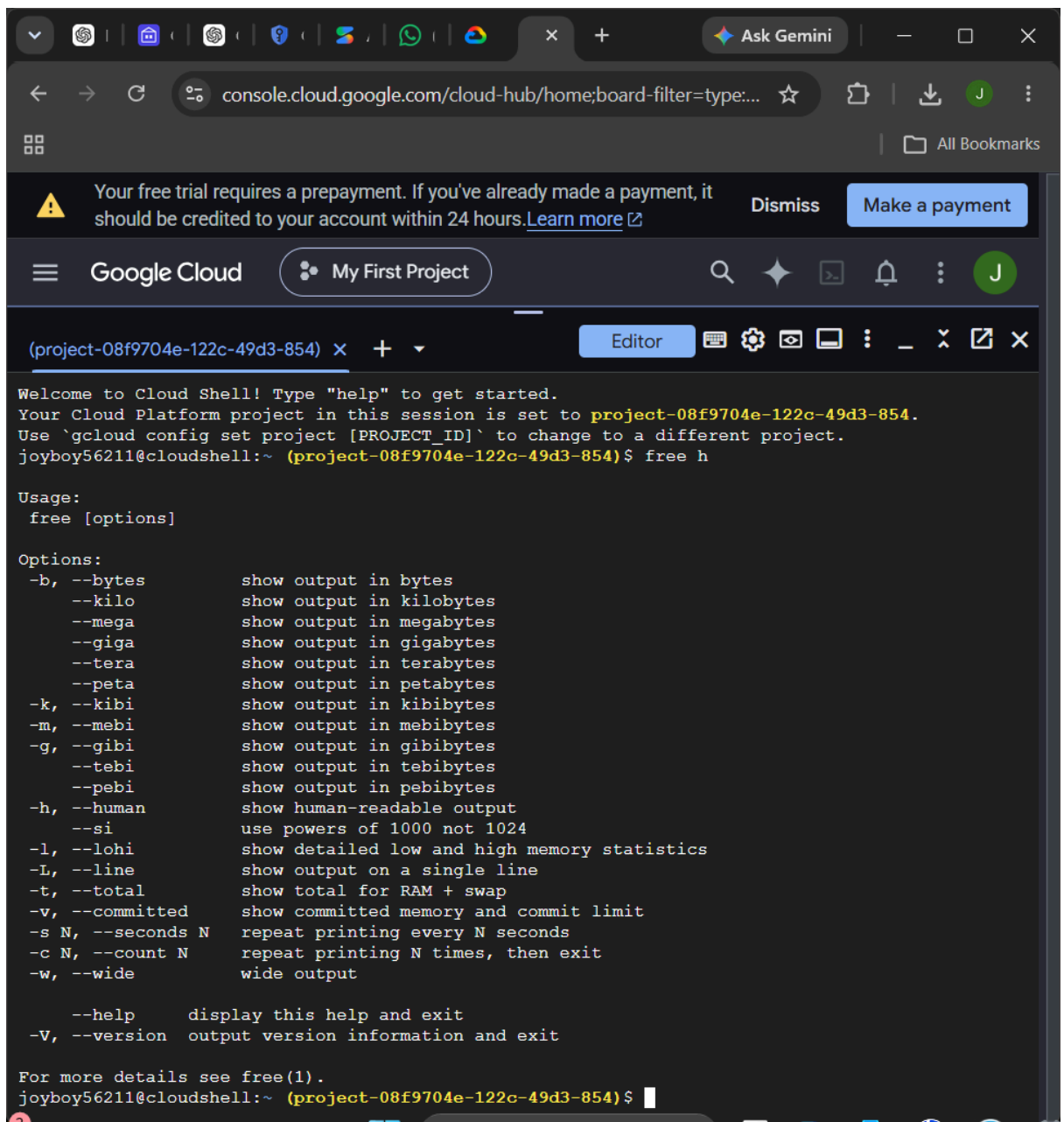
The screenshot shows the Google Cloud Shell interface. At the top, there's a navigation bar with the Google Cloud logo and 'My First Project'. Below it, a terminal window is open for project-08f9704e-122c-49d3-854. The terminal shows the following commands and output:

```
GNU nano 7.2 index.html
<html>
<body>
<h1>google cloud practical</h1>
<p>myfirst cloud web application</p>
</body>
</html>
```

At the bottom of the terminal, a file creation dialog is visible, showing the file name 'index.html' and a list of options: Help, Cancel, DOS Format, Mac Format, Append, Prepend, Backup File, and Browse.



## 4. System Resource Monitoring in Google Cloud Shell



2.

Ask Gemini

console.cloud.google.com/cloud-hub/home;board-filter=type:... ☆

All Bookmarks

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Google Cloud My First Project 🔍 ✨ > 🔔 ⋮ J

(project-08f9704e-122c-49d3-854) x + ▾ Editor 🗑 ⚙ 📺 🖨 ⋮ \_ ✕ 🗑 ✕

```
Options:
-b, --bytes          show output in bytes
--kilo               show output in kilobytes
--mega              show output in megabytes
--giga              show output in gigabytes
--tera              show output in terabytes
--peta              show output in petabytes
-k, --kibi           show output in kibibytes
-m, --mebi           show output in mebibytes
-g, --gibi           show output in gibibytes
--tebi              show output in tebibytes
--pebi              show output in pebibytes
-h, --human          show human-readable output
--si                 use powers of 1000 not 1024
-l, --lohi           show detailed low and high memory statistics
-L, --line           show output on a single line
-t, --total          show total for RAM + swap
-v, --committed     show committed memory and commit limit
-s N, --seconds N   repeat printing every N seconds
-c N, --count N     repeat printing N times, then exit
-w, --wide           wide output

--help              display this help and exit
-V, --version       output version information and exit

For more details see free(1).
```

```
joyboy56211@cloudshell:~ (project-08f9704e-122c-49d3-854)$ df -h
Filesystem      Size  Used Avail Use% Mounted on
overlay          119G   66G   54G   55% /
tmpfs            64M    0    64M    0% /dev
/dev/sda1        119G   66G   54G   55% /root
/dev/disk/by-id/google-home-part1 4.8G  5.9M  4.6G   1% /home
shm              64M    0    64M    0% /dev/shm
joyboy56211@cloudshell:~ (project-08f9704e-122c-49d3-854)$
```

```
hle avx2 smep bmi2 erms invpcid rtm rdseed adx smap xsaveopt ara
t md_clear arch_capabilities

Virtualization features:
Hypervisor vendor: KVM
Virtualization type: full
Caches (sum of all):
L1d: 32 KiB (1 instance)
L1i: 32 KiB (1 instance)
L2: 256 KiB (1 instance)
L3: 55 MiB (1 instance)
NUMA:
NUMA node(s): 1
NUMA node0 CPU(s): 0,1
Vulnerabilities:
Gather data sampling: Not affected
Indirect target selection: Mitigation; Aligned branch/return thunks
Itlb multihit: Not affected
L1tf: Mitigation; PTE Inversion
Mds: Mitigation; Clear CPU buffers; SMT Host state unknown
Meltdown: Mitigation; PTI
Mmio stale data: Vulnerable; Clear CPU buffers attempted, no microcode; SMT Host
state unknown
Reg file data sampling: Not affected
Retbleed: Mitigation; IBRS
Spec rstack overflow: Not affected
Spec store bypass: Mitigation; Speculative Store Bypass disabled via prctl
Spectre v1: Mitigation; usercopy/swapgs barriers and __user pointer sanitiza
tion
Spectre v2: Mitigation; IBRS; IBPB conditional; STIBP conditional; RSB filli
ng; PBRB-eIBRS Not affected; BHI SW loop, KVM SW loop
Srbds: Not affected
Taa: Not affected
Tsx async abort: Mitigation; Clear CPU buffers; SMT Host state unknown
Vmscape: Not affected
joyboy56211@cloudshell:~ (project-08f9704e-122c-49d3-854) $
```

```
-bash: lscpu: command not found
joyboy56211@cloudshell:~ (project-08f9704e-122c-49d3-854) $ lscpu
-bash: lscpu: command not found
joyboy56211@cloudshell:~ (project-08f9704e-122c-49d3-854) $ lscpu
Architecture: x86_64
CPU op-mode(s): 32-bit, 64-bit
Address sizes: 46 bits physical, 48 bits virtual
Byte Order: Little Endian
CPU(s): 2
On-line CPU(s) list: 0,1
Vendor ID: GenuineIntel
Model name: Intel(R) Xeon(R) CPU @ 2.20GHz
CPU family: 6
Model: 79
Thread(s) per core: 2
Core(s) per socket: 1
Socket(s): 1
Stepping: 0
BogoMIPS: 4400.36
Flags: fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pa
t pse36 clflush mmx fxsr sse sse2 ss ht syscall nx pdpe1gb rdtsc
p lm constant_tsc rep_good noopl xtopology nonstop_tsc cpuid tsc_
known_freq pni pclmulqdq ssse3 fma cx16 pcid sse4_1 sse4_2 x2api
c movbe popcnt aes xsave avx f16c rdrand hypervisor lahf_lm abm
3dnowprefetch pt1 srbds ibrs ibpb stibp fsgsbase tsc_adjust bmi1
hle avx2 smep bmi2 erms invpcid rtm rdseed adx smap xsaveopt ara
t md_clear arch_capabilities

Virtualization features:
Hypervisor vendor: KVM
Virtualization type: full
Caches (sum of all):
L1d: 32 KiB (1 instance)
L1i: 32 KiB (1 instance)
L2: 256 KiB (1 instance)
L3: 55 MiB (1 instance)
```

```
console.cloud.google.com/cloud-hub/home;board-filter=type:...

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Google Cloud My First Project

(project-08f9704e-122c-49d3-854) Editor

%Cpu(s):  0.8 us,  0.5 sy,  0.0 ni, 98.5 id,  0.0 wa,  0.0 hi,  0.2 si,  0.0 st
MiB Mem : 7947.4 total, 4847.7 free, 1820.2 used, 1523.9 buff/cache
MiB Swap:  0.0 total,  0.0 free,  0.0 used, 6127.2 avail Mem

PID USER      PR  NI  VIRT  RES  SHR  S  %CPU  %MEM    TIME+  COMMAND
 757 joyboy5+  20   -10  7360   2912  2064 S   0.3   0.0   0:00.18 tmux
1210 joyboy5+  20   0 1759316 126856 52344 S   0.3   1.6   0:10.31 MainThread
1991 joyboy5+  20   0 1662784 72816 47396 S   0.3   0.9   0:01.16 MainThread
2067 joyboy5+  20   0 1466152 72908 42900 S   0.3   0.9   0:01.20 MainThread
2474 joyboy5+  20   0 1756528 108636 52608 S   0.3   1.3   0:01.71 MainThread
   1 root      20   0  4324   3356  3076 S   0.0   0.0   0:00.07 bash
   9 syslog   20   0 222140  3852  2484 S   0.0   0.0   0:00.03 rsyslogd
  25 root     20   0 38796 28144 10912 S   0.0   0.3   0:00.91 python
  26 root     20   0  6196  2624  2380 S   0.0   0.0   0:00.00 logger
  75 root     10  -10 12020  4076  2964 S   0.0   0.1   0:00.02 sshd
  79 root     10  -10 12016  2908  1800 S   0.0   0.0   0:00.00 sshd
 406 root     20   0 2090716 87892 64580 S   0.0   1.1   0:00.90 dockerd
 501 root     20   0 1269132 13716  8552 S   0.0   0.2   0:00.97 editor-proxy
 502 root     20   0 16328  6356  5432 S   0.0   0.1   0:00.02 sudo
 506 root     20   0 1227876 3008  2164 S   0.0   0.0   0:00.00 tmux-agent
 514 root     20   0 1821204 42304 30132 S   0.0   0.5   0:02.40 containerd
 701 root     20   0  2696  1592  1496 S   0.0   0.0   0:00.00 sleep
 738 root     10  -10 13992  9680  8304 S   0.0   0.1   0:00.02 sshd
 742 joyboy5+ 10  -10 14380  6680  4896 S   0.0   0.1   0:00.01 sshd
 744 joyboy5+ 10  -10  7340  3776  3524 S   0.0   0.0   0:00.00 bash
 747 joyboy5+ 10  -10  7604  4556  3964 S   0.0   0.1   0:00.00 bash
 758 joyboy5+ 10  -10  9444  4460  4040 S   0.0   0.1   0:00.09 bash
1179 joyboy5+ 10  -10  7456  4684  3308 S   0.0   0.1   0:00.03 nano
1187 root     20   0  9196  4136  3760 S   0.0   0.1   0:00.00 runuser
1188 joyboy5+ 20   0  4324  3368  3104 S   0.0   0.0   0:00.01 sh
1233 joyboy5+ 10  -10  9444  4684  4064 S   0.0   0.1   0:00.05 bash
1960 joyboy5+ 20   0 18.4g 958432 56844 S   0.0  11.8   0:25.89 MainThread
2050 joyboy5+ 20   0 1489664 73544 48664 S   0.0   0.9   0:00.35 MainThread
joyboy56211@cloudshell:~ (project-08f9704e-122c-49d3-854) $ nano add.c
joyboy56211@cloudshell:~ (project-08f9704e-122c-49d3-854) $ gcc add.c -o add
```

## 5. Execute a C Program in Google Cloud Shell

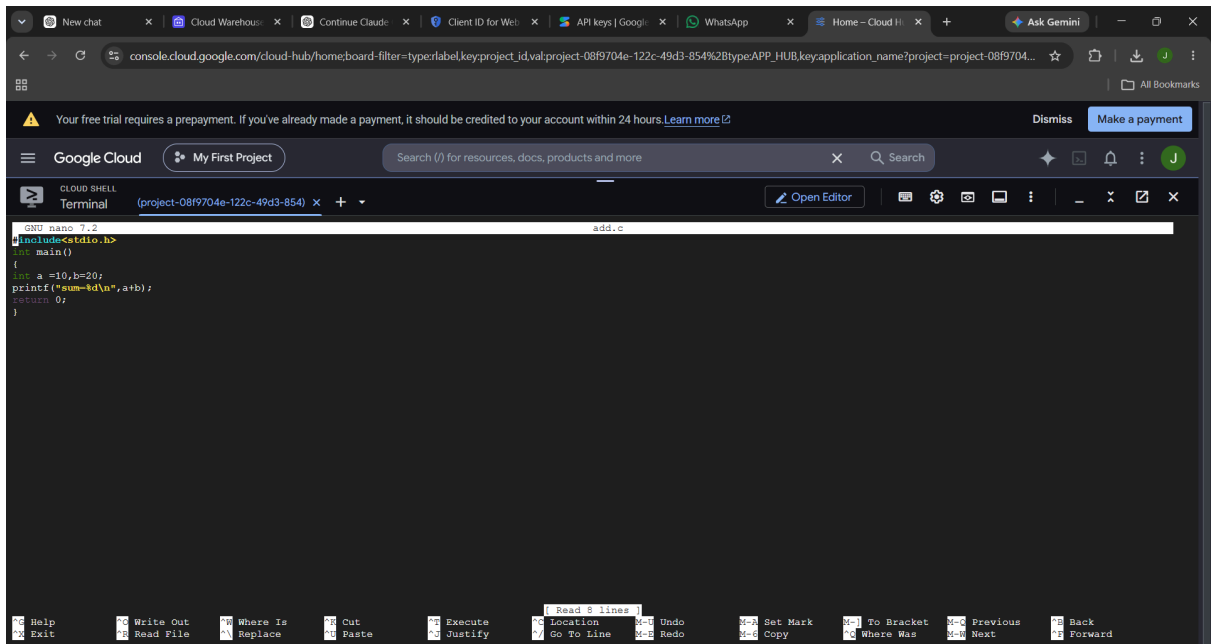
```
console.cloud.google.com/cloud-hub/home;board-filter=type:...

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Google Cloud My First Project

(project-08f9704e-122c-49d3-854) Editor

PID USER      PR  NI  VIRT  RES  SHR  S  %CPU  %MEM    TIME+  COMMAND
 757 joyboy5+  20   -10  7360   2912  2064 S   0.3   0.0   0:00.18 tmux
1210 joyboy5+  20   0 1759316 126856 52344 S   0.3   1.6   0:10.31 MainThread
1991 joyboy5+  20   0 1662784 72816 47396 S   0.3   0.9   0:01.16 MainThread
2067 joyboy5+  20   0 1466152 72908 42900 S   0.3   0.9   0:01.20 MainThread
2474 joyboy5+  20   0 1756528 108636 52608 S   0.3   1.3   0:01.71 MainThread
   1 root      20   0  4324   3356  3076 S   0.0   0.0   0:00.07 bash
   9 syslog   20   0 222140  3852  2484 S   0.0   0.0   0:00.03 rsyslogd
  25 root     20   0 38796 28144 10912 S   0.0   0.3   0:00.91 python
  26 root     20   0  6196  2624  2380 S   0.0   0.0   0:00.00 logger
  75 root     10  -10 12020  4076  2964 S   0.0   0.1   0:00.02 sshd
  79 root     10  -10 12016  2908  1800 S   0.0   0.0   0:00.00 sshd
 406 root     20   0 2090716 87892 64580 S   0.0   1.1   0:00.90 dockerd
 501 root     20   0 1269132 13716  8552 S   0.0   0.2   0:00.97 editor-proxy
 502 root     20   0 16328  6356  5432 S   0.0   0.1   0:00.02 sudo
 506 root     20   0 1227876 3008  2164 S   0.0   0.0   0:00.00 tmux-agent
 514 root     20   0 1821204 42304 30132 S   0.0   0.5   0:02.40 containerd
 701 root     20   0  2696  1592  1496 S   0.0   0.0   0:00.00 sleep
 738 root     10  -10 13992  9680  8304 S   0.0   0.1   0:00.02 sshd
 742 joyboy5+ 10  -10 14380  6680  4896 S   0.0   0.1   0:00.01 sshd
 744 joyboy5+ 10  -10  7340  3776  3524 S   0.0   0.0   0:00.00 bash
 747 joyboy5+ 10  -10  7604  4556  3964 S   0.0   0.1   0:00.00 bash
 758 joyboy5+ 10  -10  9444  4460  4040 S   0.0   0.1   0:00.09 bash
1179 joyboy5+ 10  -10  7456  4684  3308 S   0.0   0.1   0:00.03 nano
1187 root     20   0  9196  4136  3760 S   0.0   0.1   0:00.00 runuser
1188 joyboy5+ 20   0  4324  3368  3104 S   0.0   0.0   0:00.01 sh
1233 joyboy5+ 10  -10  9444  4684  4064 S   0.0   0.1   0:00.05 bash
1960 joyboy5+ 20   0 18.4g 958432 56844 S   0.0  11.8   0:25.89 MainThread
2050 joyboy5+ 20   0 1489664 73544 48664 S   0.0   0.9   0:00.35 MainThread
joyboy56211@cloudshell:~ (project-08f9704e-122c-49d3-854) $ nano add.c
joyboy56211@cloudshell:~ (project-08f9704e-122c-49d3-854) $ gcc add.c -o add
joyboy56211@cloudshell:~ (project-08f9704e-122c-49d3-854) $ ./add
sum=30
joyboy56211@cloudshell:~ (project-08f9704e-122c-49d3-854) $
```



## 6. Simple Python Web Application – PaaS

```
Cloud Shell Editor – Cloud Shell x + Ask Gemini - □ x
← → ↻ console.cloud.google.com/cloudshelleditor?refe... ☆ | J Relaunch to update
☐ | ☐ All Bookmarks
☰ Google Cloud ☰ My First Project 🔍 🔔 ⋮ J
(project-08f9704e-122c-49d3-854) x + ▾ Editor 📄 ⚙️ 🖥️ 📄 ⋮ _ ✕ 🗑️ ✕
joyboy56211@cloudshell:~ (project-08f9704e-122c-49d3-854)$ nano app.py
joyboy56211@cloudshell:~ (project-08f9704e-122c-49d3-854)$ python3 app.py
Traceback (most recent call last):
  File "/home/joyboy56211/app.py", line 7, in <module>
    HTTPServer(("", 8080), MyHandler).serve_forever()
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/lib/python3.12/socketserver.py", line 457, in __init__
    self.server_bind()
  File "/usr/lib/python3.12/http/server.py", line 136, in server_bind
    socketserver.TCPServer.server_bind(self)
  File "/usr/lib/python3.12/socketserver.py", line 473, in server_bind
    self.socket.bind(self.server_address)
OSError: [Errno 98] Address already in use
joyboy56211@cloudshell:~ (project-08f9704e-122c-49d3-854)$ nano app.py
joyboy56211@cloudshell:~ (project-08f9704e-122c-49d3-854)$ python3 app.py
Traceback (most recent call last):
  File "/home/joyboy56211/app.py", line 7, in <module>
    HTTPServer(("", 8080), MyHandler).serve_forever()
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/lib/python3.12/socketserver.py", line 457, in __init__
    self.server_bind()
  File "/usr/lib/python3.12/http/server.py", line 136, in server_bind
    socketserver.TCPServer.server_bind(self)
  File "/usr/lib/python3.12/socketserver.py", line 473, in server_bind
    self.socket.bind(self.server_address)
OSError: [Errno 98] Address already in use
joyboy56211@cloudshell:~ (project-08f9704e-122c-49d3-854)$ mkdir storage_demo
joyboy56211@cloudshell:~ (project-08f9704e-122c-49d3-854)$ cd storage_demo
joyboy56211@cloudshell:~/storage_demo (project-08f9704e-122c-49d3-854)$ echo "welcome to google cloud storage demo">student.txt
-bash: echowelcome to google cloud storage demo: command not found
joyboy56211@cloudshell:~/storage_demo (project-08f9704e-122c-49d3-854)$ cat student.txt
joyboy56211@cloudshell:~/storage_demo (project-08f9704e-122c-49d3-854)$ echo "welcome to google cloud storage demo">student.txt
joyboy56211@cloudshell:~/storage_demo (project-08f9704e-122c-49d3-854)$ cat student.txt
welcome to google cloud storage demo
joyboy56211@cloudshell:~/storage_demo (project-08f9704e-122c-49d3-854)$ python3 -m http.server 8080
```